$\boxed{2\text{-zero}}$

Ⓐ 24, 15, 16, 17, ---, 1 n-1

Ⓑ 2n, 35, 36, 37 --- 3 n-1

Ⓒ from each $(1\ell, 2\ell, 3\ell)$ $\ell = 5, --, n-1$ pick one.

$\boxed{3\text{-zero}}$

$$X_{46} = X_{16} - X_{15}$$
$$X_{47} = X_{17} - X_{15}$$
$$\vdots$$
$$X_{4\,n-1} = X_{1\,n-1} - X_{15}$$
$$X_{4n} = -X_{15}$$

$$\left.\begin{aligned} X_{2j} &= X_{4j} + X_{25} \\ X_{3j} &= X_{4j} + X_{35} \end{aligned}\right\} \quad j = 6, --, n$$

$$X_{2j} = X_{1j} + X_{2n}$$
$$X_{3j} = X_{1j} + X_{3n} \qquad j = 5, --, n-1$$

- if X_{25} finite : must have $X_{4j} \to \infty$, $X_{2j} \to \infty$ $j = 6, --, n$
- if X_{35} finite : must have $X_{4j} \to \infty$, $X_{3j} \to \infty$ $j = 6, --, n$
- if X_{2n} finite: must have $X_{1j} \to \infty$, $X_{2j} \to \infty$ $j = 5, --, n-1$
- if X_{3n} finite: must have $X_{1j} \to \infty$, $X_{3j} \to \infty$ $j = 5, --, n-1$
- if X_{2j} $(j = 6, --, n-1)$ is finite then $X_{2j} = X_{4j} + X_{25}$ implies that must have $|X_{25}| \to \infty$ $|X_{4j}| \to \infty$
  also $X_{2j} = X_{1j} + X_{2n}$ so $|X_{2n}| \to \infty$, $|X_{1j}| \to \infty$
- if X_{3j} $(j = 6, --, n-1)$ is finite then $|X_{35}|, |X_{3n}|, |X_{1j}|, |X_{4j}| \to \infty$

$$X_{2j} = X_{4j} + X_{25}$$
$$X_{3j} = X_{4j} + X_{35}$$
$$X_{2j} = X_{1j} + X_{2n}$$
$$X_{4n} = -X_{15}$$

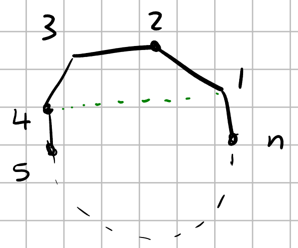
k-zero

Ⓐ 2 k+2, 3 k+2, --, k k+2
1 k+3, 1 k+4, ---, 1 n-1 free

Ⓑ 2n, 3n, ---, kn
k+1 k+3, k+1 k+4, ---, k+1 n-1 free

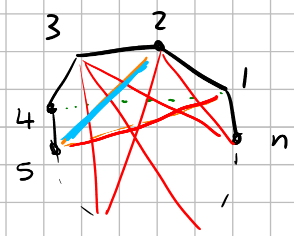
Ⓒ for each $\ell \in \{k+3, --, n-1\}$ pick one of $\left\{\begin{array}{l} X_{1\ell} \\ X_{k+1\,\ell} \\ \text{all of } X_{2\ell}, X_{3\ell}, --, X_{k\ell} \end{array}\right.$

<u>what survives?</u>

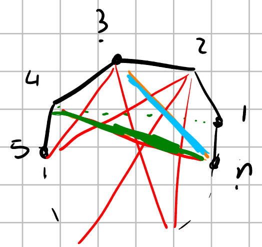


if nothing crosses, we are in the old case, so assume crossing.

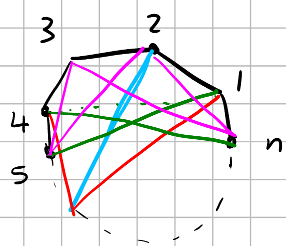
- if we have X_{25} or X_{35} it survives if one of the following happens:
  - any other X_{2j} $(j=6,\dots,n)$ is present
  - any other X_{3j} $(j=6,\dots,n)$ is present
  - any of X_{4j} $(j=6,\dots,n)$ is present
  - or X_{15} is present



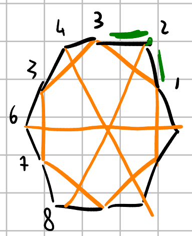
- if we have X_{2n} or X_{3n} it survives if one of the following happens:
  - any other X_{2j} $(j=5,\dots,n-1)$ is present
  - any other X_{3j} $(j=5,\dots,n-1)$ is present
  - any of X_{1j} $(j=5,\dots,n-1)$ is present
  - or X_{4n} is present



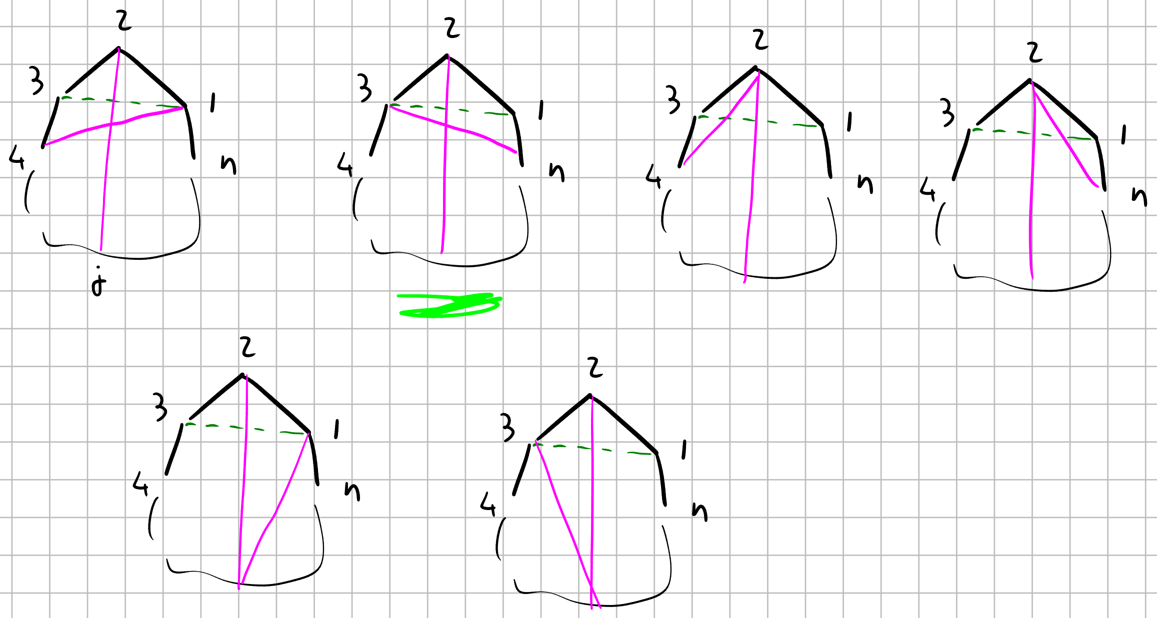
- if we have $X_{2\ell}$ or $X_{3\ell}$ $(\ell=6,\dots,n-1)$ it survives if one of the following happens:
  - X_{15} or X_{4n} are present
  - either $X_{1\ell}$ or $X_{4\ell}$ are present
  - either X_{25} or X_{35} are present
  - either X_{2n} or X_{3n} are present



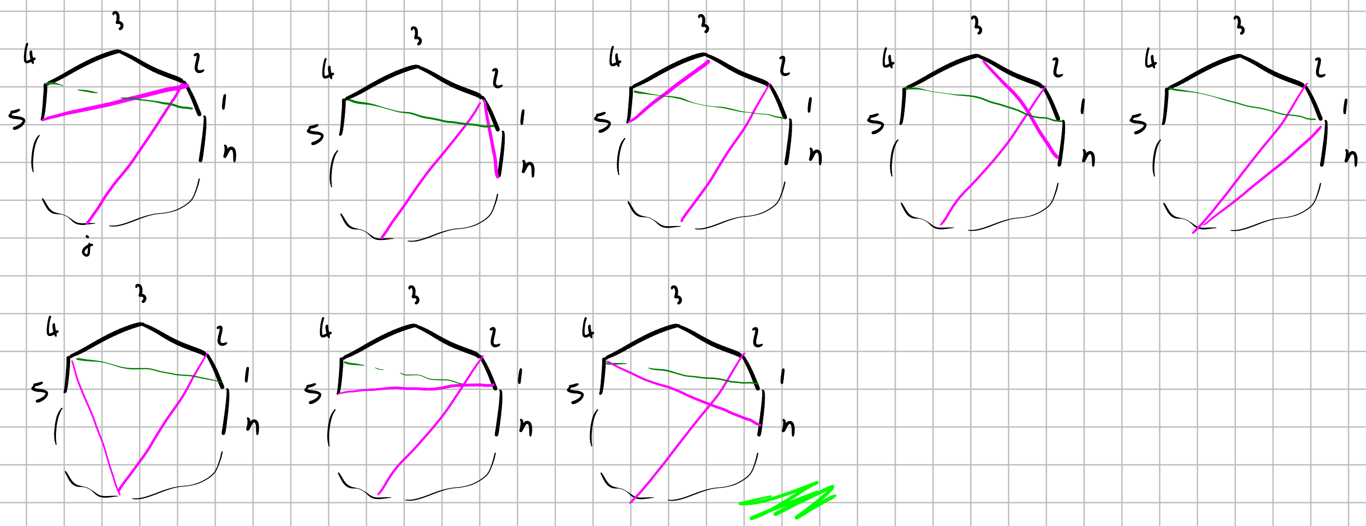
(c) For each $\ell \in \{k+3,\dots,n-1\}$ pick one of $\begin{cases} X_{1\ell} \\ X_{k+1\,\ell} \\ \text{all of } X_{2\ell}, X_{3\ell}, \dots, X_{k\ell} \end{cases}$



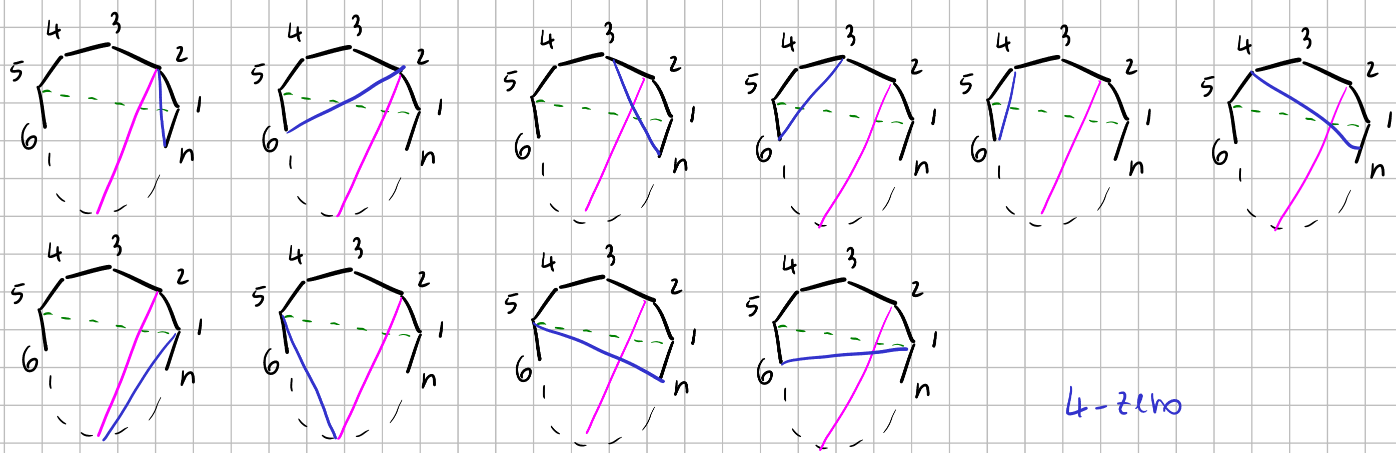
$k=2$ wins the prize (point (c))



© for 3-zero



$X_{n-2,n}$ is born free for all k-zeros starting at 1 (idea for future)



4-zero